

Integrating new variables into a framework to support cacao denomination of origin: a case study in Southwest Colombia

Carlos E González-Orozco,^{a,*} Mario Porcel,^{a,b} Roxana Yockteng,^{c,d} Alejandro Caro-Quintero,^{c,e} Caren Rodriguez-Medina,^{f,g} Margareth Santander,^{c,g} Martha Zuluaga,^c Mauricio Soto,^c Jader Rodriguez Cortina,^c Fabrice Eric Vaillant^f and Sebastian Escobar Parra^{f,g,*}



Abstract

BACKGROUND: Cocoa quality plays a pivotal role in establishing denominations of origin, with genotypes, geography, climate and soil conditions being key variables. However, these factors have not been comprehensively explored in defining cacao denominations of origin. The present study addresses this gap by laying the foundation for cacao denomination of origin, focusing on the Buenaventura region on Colombia's Pacific coast. Our goal is to provide a holistic understanding of the elements underpinning cacao denomination of origin, emphasizing Buenaventura's unique cocoa quality and geographical significance.

RESULTS: Through the Buenaventura case, we propose a robust framework applicable to other cacao-producing regions, elevating the recognition and value of cacao denomination of origin. Our framework encompasses geography, agronomy, genetics, microbial diversity, pests and diseases and cocoa quality. In a pioneering move, we propose a cacao denomination of origin in Colombia, specifically examining Bajo Calima, Sabaletas and Cisneros within Buenaventura region. Buenaventura stands out for its cocoa quality, characterized by fruity flavors attributed to the rich biodiversity of the lowland rainforest.

CONCLUSION: Our analysis indicates specific geographical indicators for each of the study zones, with Buenaventura identified as a region with natural characteristics to produce fine flavour cocoa products. Each zone exhibited a high differentiation and diversity of cacao cultivars. Buenaventura has the potential to be designated as a future denomination of origin for cacao from the Pacific region of Colombia, characterized by its unique fruity-aroma chocolates. Our framework is adaptable to other cacao-producing regions, facilitating the establishment of denominations of origin within the cocoa industry and agriculture.

© 2023 The Authors. *Journal of The Science of Food and Agriculture* published by John Wiley & Sons Ltd on behalf of Society of Chemical Industry.

Supporting information may be found in the online version of this article.

Keywords: Buenaventura; chocolate; cocoa flavour-aroma; rainforest

* Correspondence to: C E González-Orozco, Corporación Colombiana de Investigación Agropecuaria – Agrosavia, Centro de Investigación La Libertad- Km 14 vía Villavicencio-Puerto López, Meta, Colombia. E-mail: cegonzalez@agrosavia.co; or S Escobar, E-mail: sescobar@agrosavia.co

a Corporación Colombiana de Investigación Agropecuaria – Agrosavia, Centro de Investigación La Libertad, km 14 vía Puerto Lopez, VILLAVICENCIO, Meta, Colombia

b Instituto de Investigación y Formación Agraria, Pesquera, Alimentaria y de la Producción Ecológica (IFAPA), Málaga, Spain

c Corporación Colombiana de Investigación Agropecuaria – Agrosavia, Centro de Investigación Tibaitatá, vía a Mosquera, Bogotá, Colombia

d Departamento de Biología, Facultad de Ciencias, Universidad Nacional de Colombia sede Bogotá, Ciudad Universitaria, Bogotá, Colombia

e Corporación Colombiana de Investigación Agropecuaria – Agrosavia, Centro de Investigación La Selva, vía Rionegro - Las Palmas, Sector Llano Grande, Rionegro, Colombia

f Muséum National d'Histoire Naturelle, UMR-CNRS 7205, Paris, France

g Corporación Colombiana de Investigación Agropecuaria – Agrosavia, Centro de Investigación Palmira, Valle del Cauca, Colombia

INTRODUCTION

Geographical indications (GI) and denomination of origin (DO) are widely used concepts to determine the geographical and commercial status of food production.¹ According to the World Intellectual Property Organization,² a GI is a sign used on goods with a specific geographical origin and possessing qualities or a reputation because of that origin. DO, on the other hand, refers to food products specific to a particular region or town, conveying a special quality or characteristic of the designated area (European Union Commission). Although the guidelines for GI and DO are globally accepted, standardized scientific frameworks to support their application to emerging products in developing economies are lacking. Consequently, evidence-based frameworks are needed to provide rigor in defining or understanding GI and DO. The present study presents a new approach to define and support the establishment of cacao GI and DO in an understudied tropical region. We propose a science-based framework to support GI and DO establishment, using the cacao-producing Pacific region of Buenaventura in Colombia as a case study. Here, we will refer to the cacao tree as 'cacao' and derived products made from cocoa beans as 'cocoa'.

GI and DO are effective tools for differentiating foods and agro-nomical products.³ Global guidelines on GI regulations have been developed by FAO to strengthen capacities at various levels of the food industry. Even within Europe, each country develops its own regulations. In 2017, France published a booklet with user guidelines for GI and DO,⁴ providing clarity on rules for licensing DOs. In Latin America, the commission of the Andean community created a policy document on industrial property called Decision 486, which is related to patents of genetic resources. However, it does not offer a detailed guideline on DO requirements specifically; instead, it states that registration is a matter for each country. Consequently, regional legislations are unclear, complicating the application process for GIs or DOs. This lack of clarity leads to misinformation and a poor understanding of the commercial value of GIs and DOs as tools. Colombia, for example, is no exception. The Colombian Commerce and Industry institution (CCI), the government entity responsible for issuing DOs, provides just four bullet points outlining the requirements for applying for a certificate of denomination of origin. This list is clearly insufficient for the extensive task at hand in the cocoa industry.

Native tropical cacao trees and their cultivated cousins are significant sources of cocoa aromas.⁵ However, they are not utilized as frequently as expected for proposing DOs and GIs. Several examples of cacao DOs or GIs exist in South America, such as Cacao Chuau in Venezuela,⁶ Cacao Arriba in Ecuador,^{7,8} Cacao Grijalva in Mexico⁹ and Cacao Tomé-açu in Brazil.¹⁰ The Cacao Arriba region in Ecuador shares similar latitudes and climates with the Buenaventura region. In Ecuador, the provinces of Los Rios and Guayas have a long history as cacao-producing areas since the colonial period. In 2011, the Ecuadorian Institute of Intellectual Property granted a DO for Ecuador's fine aroma cacao, including the Cacao Arriba region on the Pacific Coast. However, the DO for Ecuador is applied under a broad scheme, and more specificity is needed in delimitating DO regions. For example, it would be an improvement if Ecuador could derive from the National cacao a set of specific DO's delimited based on new surveys or methodologies. In the case of Venezuela, the Cacao Chuau in Venezuela's Aragua state covers a specific geographic valley along a river that leads to the Caribbean Ocean.

Validating DOs over time is a constant challenge for these regions because Chuau's DO is still under scrutiny, and finalizing its designation remains an ongoing task. Despite a few examples

of Cacao DOs in the tropical region of South America, a lack of scientific rigor in the processes involved in supporting DO assignments is a common characteristic of all the mentioned cacao cases. The documents supporting the request for the DO license often consist of four sections: (1) a legal framework; (2) a description of the geographic region; (3) quality characteristics of the product and its marketplace; and (4) communities and cacao associativity or trading networks. Adopting a more evidence-based and focused approach would greatly benefit intellectual property and legal processes in the domain of food product qualifications.

Colombia lags behind in the realm of cacao DOs. However, it holds the potential to become a future stronghold for cacao in the region as a result of its unique geographical conditions and diverse environments, encompassing 54 cacao regions in Colombia.¹¹ As mentioned before,¹² achieving a comprehensive system to improve GI and DOs legislation in Colombia requires significant inter-institutional coordination between the government, communities and private industry. Indeed, over the last decade, the CCI has only granted 19.¹³ There are some examples of DOs in the south American coffee industry including Colombia and Honduras.^{14,15} However, the lack of scientific advances in cacao DOs and GIs in Colombia is counterproductive, considering the country's abundant agrobiodiversity, and its status as a centre of cacao origin, accompanied by cultural richness. Together, these factors present numerous opportunities in the food industry. Despite these natural advantages, Colombia has yet to issue any cacao DOs or GIs. As mentioned in Section 3.3 of the CCI technical manual, one of the weaknesses is limited scientific research on the topic, compounded by a lack of qualified professionals in the field. However, the situation is currently evolving in Colombia. Recent cacao studies provide new evidence to advance the creation of GIs for cacao in the country, such as proposing a cacao regionalization.¹¹ This regionalization is the first of its kind and serves as the scientific foundation for defining the separation of geographical cacao regions, which is useful because it goes beyond political boundaries. It represents a novel approach because the territory's geography drives the proposed classification the river systems at different levels (i.e. catchment, sub-catchment), with microclimate based on the soil temperature and macroclimate data being used to define the cacao regions. This stands in contrast to existing cacao crop classifications, which rely on agronomical aptitude and departmental classification.¹⁶

The knowledge of cacao bean flavor quality serves as an example to illustrate the differentiation of cacao origins.¹⁷ Similar to Venezuela and Ecuador, Colombia also produces fine-flavored cocoa. Certain Colombian regions have gained worldwide recognition for their high-quality cocoa, solely based on organoleptic characteristics. However, there is a lack of scientific data supporting the holistic evaluation of Colombian cacao beans. Nevertheless, recent studies have demonstrated the unique sensory qualities of Colombian cacao.^{18,19} Cocoa beans from the Arauca region in eastern Colombia have twice been awarded the title of best cacao in the world at the 'International Cacao Awards' held during the 'Salon du Chocolat' in Paris. This recognition was also achieved by cacao beans from Tumaco region, situated in southwestern Colombia geographically close to the Buenaventura region. Additionally, other studies emphasized the quality potential of the FEAR 5 cacao cultivar developed in the Arauca region, known by its fruity and nutty aromas.¹⁹ In parallel, more work has been carried out on developing a fermentation methodology for cacao cultivated in the pacific region,¹⁸ aiming to enable cacao farmers to consistently produce fine flavored cacao beans. These

are examples of cultivated cacao, but Colombia is home to many unexplored materials of wild cacao species,²⁰ with numerous wild genotypes from the Amazon still waiting to be discovered and utilized in determining future designations of origin.

To date, no comprehensive characterization has been conducted to define or classify cocoa quality based on its genetic diversity, regional or local climate, soil characteristics, landscape features, sensorial and biochemical indicators, and productivity aspects. This holds true not only for Colombia, but also for other cacao-producing countries in South America. Consequently, the series of steps and variables that our framework contains represent a significant advancement in the field of cacao DOs and GIs, particularly for unexplored geographic areas within the tropical Pacific region of Colombia. The present study demonstrates the feasibility of establishing cacao DOs and GIs by employing a multidisciplinary and robust scientific approach that lays the foundation for supporting cacao DOs through a series of independent steps.

MATERIALS AND METHODS

Study region and sampling zones

The Buenaventura region is located in the department of Valle del Cauca, situated in the southwestern part of Colombia along the Pacific Ocean (Fig. 1). Buenaventura is part of the Biogeographical Chocó region.²¹ The region is encompassed by tropical rainforests that experiences some of the highest precipitation levels on earth.²² The exceptionally high rainfall is a result of the convergence of vast amounts of air rising from the ocean, combined with the intricate and rugged topography of the area, as well as the intense evapotranspiration occurring within the tropical rainforest. Within the Buenaventura region, three specific cacao sampling zones were selected, each representing a distinct cacao cropping system.

The average climate conditions for each study zone are presented in Table 1. Zone 1 (Cisneros) is situated closer to the foothills of the western Andean mountain range at an elevation of 515 m a.s.l. It experiences the highest amount of solar radiation and notable temperature variations between the maximum and minimum values during the day and night. Zone 2 (Bajo Calima), located on the lowlands closer to the Pacific Ocean at approximately 40 m a.s.l, is characterized as the rainiest and most humid among all three zones. Lastly, zone 3 (Sabaletas) is positioned further south on the marine plains of the coastal areas, with an elevation of 15 m a.s.l. This zone exhibits the highest average maximum temperature and is the warmest among the three.

The soils found in the cacao plantations of the study region are primarily Inceptisols and Entisols. These soils are characteristic of the area and are highly acidic, with pH levels from 5.0 to 5.4. Additionally, they occasionally show significant limitations as a result of aluminium toxicity. Lime application and fertilization practices are infrequent. The dominant soil textures in the region are loam and clay loam, and the organic matter content varies, typically ranging from moderate to high levels.

Summary of the proposed framework

Our framework is not a mere variable-based methodology but comprises a series of steps to establish cacao denominations of origin in tropical regions. Cocoa quality, including other unique traits, is vital for such denominations. Our proposal integrates these variables, contributing significantly to the concept,

although we do not claim to devise a comprehensive analytical methodology.

The proposed bottom-up framework comprises two interconnected cycles (green and blue cycles in Fig. 2). The first step is a regionalization mapping, which involves scaling from a national to regional level using climate and soil indicators specific to each geographic area (green cycle). Once the region of interest is identified, an agronomical assessment is conducted to characterize the cacao cropping system, including the evaluation of pests and diseases present in the region. Moving forward, the blue cycle focuses on quantifying product typicity variables of cacao at both the individual farm and clusters of farms levels. This cycle involves conducting a genetic identity analysis, a detailed typification of cocoa quality, and a survey of microorganism diversity, all of which are crucial in defining the sensorial characteristics of chocolate. All these steps are interconnected and collectively constitute the fundamentals and key variables in the proposed framework that are required to achieve a well-established denomination of origin for cacao.

Geography

Regionalization of GIs

A biogeographical regionalization approach was employed to identify distinct cacao regions in Colombia, which were subsequently designated as GIs.¹¹ To conduct the regionalization, a dataset consisting of 4974 cacao farm occurrences of cacao across Colombia, along with macroclimate variables,²³ and microclimate variables,²⁴ were utilized as inputs. The macroclimate WorldClim v.2 dataset²³ comprises seven variables, including precipitation, maximum temperature, minimum temperature, mean temperature, relative humidity, wind speed and solar radiation, with a spatial resolution of 1 km² and a soil depth of 0–5 cm. The microclimate analysis employed the SBIO dataset,²⁴ which contains eleven soil temperature variables. The macroclimate analysis enabled an understanding of the climate variations between different geographic regions, whereas the microclimate analysis focused on determining specific microclimate conditions based on soil properties.

Mean climate and soil temperature values for each cacao farm occurrence were extracted from the WorldClim2 climate raster and the SBIO soil temperature raster. Principal component analysis (PCA) was performed using the extracted climate data from WorldClim2 and SBIO rasters, employing R, version 3.6.3 (R Foundation, Vienna, Austria) and the ‘vegan’ package. The PCA axis 1 results were then mapped onto the climate and soil temperature values using a geospatial application.²⁵ This application facilitated the generation of climate and soil temperature maps for all cacao producing regions in Colombia, based on the principal components. These results serve as GIs, reflecting specific geographical origins, and were utilized to identify the cacao GI for Buenaventura at the national level. Lastly, a geospatial local-level PCA analysis was independently conducted for the three sampling zones using the same methodology, aiming to characterize the finer differences in local climate and soils between the three zones. This local analysis can be considered as a fine-scale identification of GIs.

Agronomic baseline

A technical diagnosis of cacao farmers was conducted using a participatory methodology, where farmers provided descriptive information regarding their agronomic practices and farming systems. In total, 12 cacao farms for the three zones were visited, and

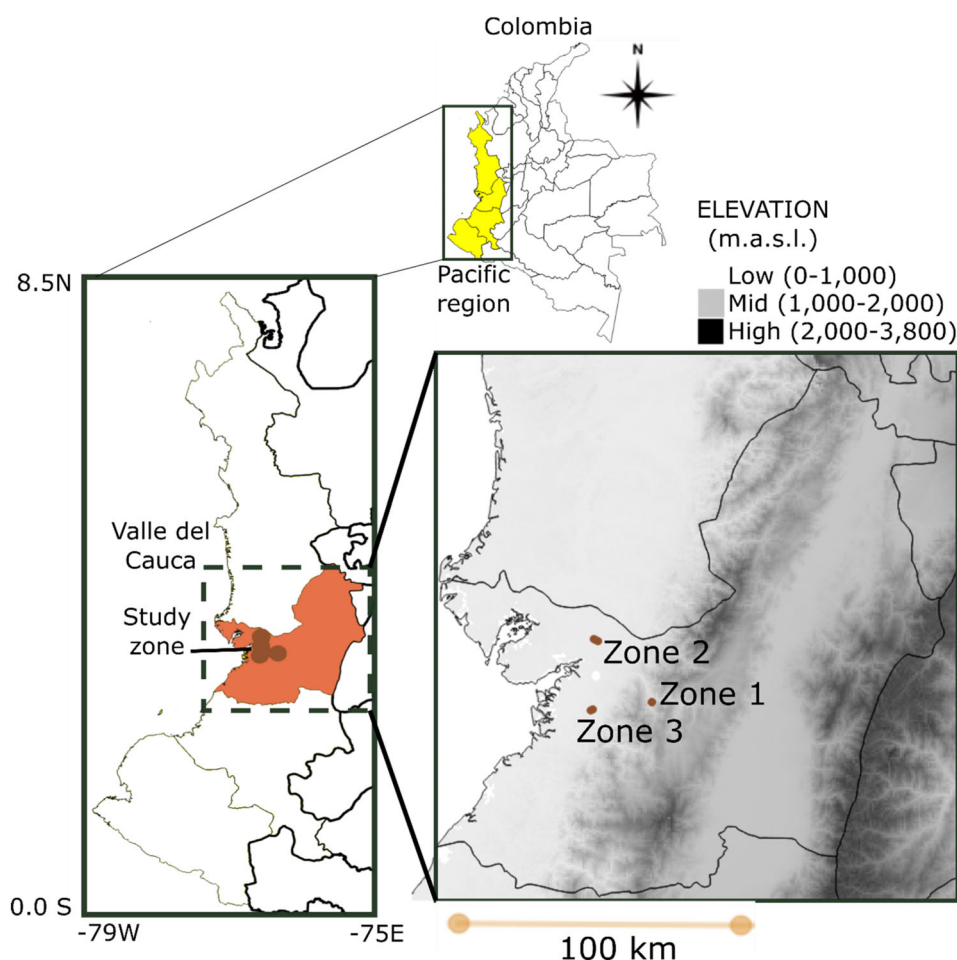


Figure 1. Map of Colombia with the location of the Buenaventura study region on the Pacific coast and the three sampling zones.

Table 1. Average climate conditions for the three sampling zones

Zones	Diurnal temp range (°C)	Solar radiation (kJ m ⁻² day ⁻¹)	Maximum temperature (°C)	Mean temperature (°C)	Minimum temperature (°C)	Rainfall (mm)	Vapor pressure (kPa)	Wind speed (m s ⁻¹)
1. Cisneros	<u>8.74</u>	<u>16.310</u>	29.06	24.70	<u>20.32</u>	2686	2.52	1.05
2. Bajo Calima	7.65	15.766	30.15	26.33	22.45	<u>6262</u>	<u>2.88</u>	<u>1.31</u>
3. Sabaletas	8.25	16.123	<u>30.54</u>	<u>26.37</u>	22.24	4169	2.86	1.21

Note: Underlined values are the maximum averages of the three zones for each climate variable.

interviews were conducted with 32 farmers. Some of these farmers were members of cacao grower associations active in the study area, including ACABC (association of farmers and cacao growers of Bajo Calima), APCC (Cisneros growers association) and Sabaletas Community Council. The cacao plantations in the study region contained genetic materials of cacao related to FQ-1, FEAR5, F61, SUI-83, SUI-62, GS, TSH-565, ICS-95, CCN-51, FCH-8, FQ-1 and FEE-2.

Pests and diseases

The infestation level of the cacao pest species and the incidence of fruit diseases were determined through a participatory

approach that involved close collaboration between farmers and researchers. In total, 11 plantations were selected, with three located in zone 1, three in zone 2 and five in zone 3. The sampling took place between the 6th and 10th of October 2018, during the peak cacao harvest period of the year. In each plantation, 10 trees were randomly chosen by following a zigzag transect pattern that aimed to cover the entire field. The path and distance of the transect were approximately determined beforehand during an inspection of the plantation, with guidance from the farmer.

For each sampled tree, we conducted a comprehensive assessment of cacao fruit damage. Specifically, we recorded the damage caused by different categories of pests, including:

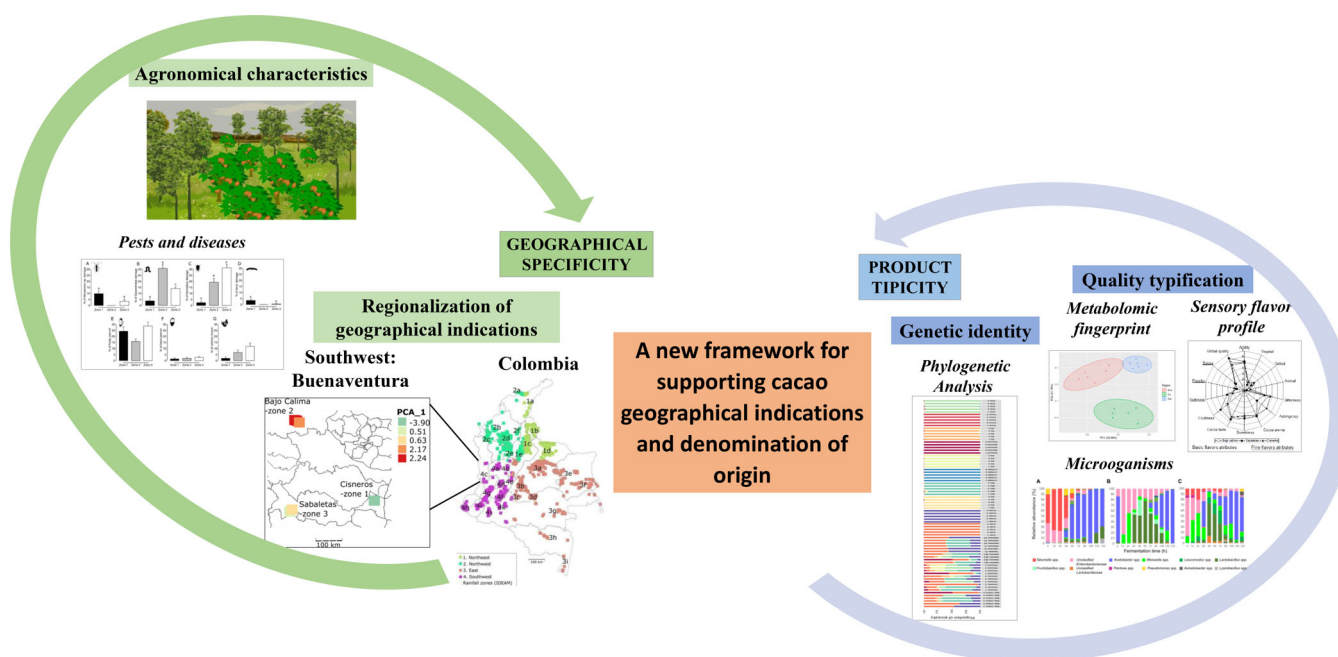


Figure 2. Flow diagram of the proposed framework.

- (1) cacao mirids (Hemiptera: Miridae),
- (2) sucking insects (other than mirids), that included treehoppers and thorn hoppers (Hemiptera: Membracidae) as well as scale insects (Hemiptera: Coccidae) and mealybugs (Hemiptera: Pseudococcidae),
- (3) pod borers (Lepidoptera),
- (4) geometer moths (Lepidoptera: Geometridae), and
- (5) rodents and other mammalians.

To identify cacao mirid damage, we looked for the characteristic dark, round spots on the pod surface²⁶ (see Supporting information, Fig. S1A). Damage caused by other pod-sucking insects was differentiated from mirid damage by its black to brown scarring, irregular shape and often connected scars, indicating the previous presence of clustered mealybug or membracid colonies (see Supporting information, Fig. S1B). These scars sometimes had yellowish–green halos, which were not observed around mirid spots. To confirm this type of damage, we removed mealybug and membracid colonies from the pod surfaces to examine the scarring caused by their feeding activity. Because of their similar appearance, this type of damage was grouped into a single category and could not be distinguished without the presence of the hemipteran colony.

Pod borer damage was characterized by black entry spots accompanied by humid frass buildup in their vicinity, often with multiple spots per pod. We checked these spots for tunneling activity and the presence of larvae. Geometrid damage (see Supporting information, Fig. S1C) was identified by cherelles or pods with gnawed scars, resulting in wide depressions or holes on the smooth surface (the latter likely caused by late instar larvae). Damage to cherelles and young fruits often led to misshapen fruits as they matured (see Supporting information, Fig. S1C). Geometrids were observed on leaves, fruits and leaf surfaces, where they consumed the plant material.

Finally, rodents created large holes in the pods to access and consume the pulp and seeds (see Supporting information,

Fig. S1E). When available, insect pests associated with the damaged pods were collected for taxonomic identification.

The primary disease issues affecting cacao, including (1) the frosty pod rot (FPR), caused by *Moniliophthora roreri* (Cif. & Par.); (2) the black-pod rot (BPR), caused by *Phytophthora* spp.; and (3) witches' broom disease (WBD), caused by *Moniliophthora perniciosa*, were identified, although not all were surveyed in the present study.

To determine the causal agent responsible for each disease, plant tissue samples were collected from infected plants. Diseased pods were carefully removed, including healthy tissues extending approximately 4 cm beyond the lesions. These samples were then placed in plastic bags and transported to the laboratory for further analysis.

In the laboratory, the samples underwent surface sterilization using a 2% sodium hypochlorite solution for 15 min. Subsequently, they were treated with 70% ethanol for 1 min and rinsed three times with sterile water. The plant tissue was dried and then sectioned into six segments measuring 0.5 cm each. These segments were directly placed on Potato Dextrose Agar (PDA; Oxoid, Basingstoke, UK) supplemented with 0.1% Triton 100X (Sigma-Aldrich, St Louis, MO, USA) and 0.3% chloramphenicol to prevent bacterial contamination. The plates were incubated at $25 \pm 2^\circ\text{C}$ under 24 h of light for 7 days. Any fungal growth observed was sub-cultured onto fresh PDA or water agar (1% w/v) until pure colonies were obtained.

To compare infestation levels across the different study zones, we employed generalized mixed effect models (GLMM) with a Poisson error distribution and a log-link, utilizing the 'lmer4' package. The experimental unit was defined as the individual tree, with the number of pods displaying signs of pest and disease damage compared to the total number of pods per tree. This ratio was used as an offset variable.²⁷ The sampling zone was included as an explanatory variable, and the plantation was treated as a normally distributed random effect to account for any spatial autocorrelation among trees within the same

transect. Pairwise post-hoc comparisons between zones were conducted using Tukey's tests on the model estimations. Models were assessed for overdispersion, and the Pearson residuals *versus* the fitted values were visually inspected to validate the models.

Genetics

Plant material and DNA extraction

Leaf samples were collected from the three study zones to obtain genotypic information on the cultivated cacao trees in the region. Leaves were sampled from ten farms in zone 1, six farms in zone 2 and 10 farms in zone 3. DNA extraction from the leaves and subsequent genotyping was performed via LGC Biosearch Technologies (Teddington, UK). KASP genotyping assays were utilized to identify single nucleotide polymorphisms (SNPs) at specific loci of the cacao genome, which had been previously used in various studies.^{28–32} Bi-allelic discrimination was achieved by competitively binding two allele-specific forward primers, each having a unique tail sequence that corresponded with two universal fluorescence resonant energy transfer cassettes. One cassette was labeled with FAM dye, whereas the other was labeled with HEX dye.

Phylogenetic analysis

Out of the initial 96-SNP panel selected for studying genetic diversity, 87 SNP markers were utilized to align the new dataset generated in the present study with the dataset published previously.³¹ The alignment of the SNP dataset was performed using ClustalX, version 1.83,³³ and visually confirmed using Geneious, version 8.1.7.³⁴ After concatenating the data, a phylogenetic tree was reconstructed using the maximum likelihood method with PhyML, version 3.0,³⁵ implemented in a bioinformatics platform (<http://www.atgc-montpellier.fr/phyml>). The phylogenetic analysis involved 100 bootstrap replicates, and gap sites were treated as missing data. The species *Theobroma grandiflorum* (Wild Ex. Spreng. Schum) was used as an outgroup in the phylogenetic tree.

To assess possible significant genetic differences between the study zones, a one-way permutational multivariate analysis of variance (PERMANOVA) was conducted, and the data was visualized through PCA using Past, version 4.1.³⁶ The PCA analysis was based on the sum of SNP differences between samples. Furthermore, the ancestry of each genotype was analyzed using the intersected SNPs representing the 10 recognized cacao genetic groups.³⁷ A maximum likelihood analysis was performed using Admixture, version 1.3,³⁸ in a supervised mode, where the reference genetic groups were specified.

Quality

Typification

One post-harvest transformation site for each zone was conducted to evaluate the quality potential of cacao cultivated. Figure 3 illustrates the stages involved in the process. The assessment of quality potential involved studying the flavour profiles of chocolate (or cocoa) and analyzing the cocoa metabolome after post-harvest processing. The entire process was carried out at a single site in each region, utilizing the same fermentation methodology and equipment.

Postharvest transformation process

Harvesting involved selecting a representative sample of cacao cultivars from each of the 32 farms in each study zone. Healthy and fully ripe cacao pods were carefully harvested one day prior to the fermentation process and then transported to the central postharvest transformation site.

The fermentation and drying processes were conducted as follows: Cacao pods were carefully opened using sharp knives, and the seeds were manually extracted and mixed to create an initial mass comprising all the collected cacao cultivars. From this initial mass, 50 kg of seeds were extracted and placed into six separate compartments within a wooden fermentation box (Fig. 3), allowing for six repetitions of the process. The fermentation followed a standard method widely employed by cacao producers in Colombia.³⁹ Initially, the cacao mass underwent an anaerobic stage lasting 48 h without any mixing. Subsequently, mixing and aeration of the cacao seeds were performed every 12 h for a total duration of 120 h. Throughout the fermentation process, the temperature of the fermentation mass was monitored at the central point of each compartment using a thermocouple (model 735-1; Titisee-Neustadt, Germany). Finally, the cacao samples intended for sensory and metabolomics analysis were dried in a forced convection oven at 50 °C, utilizing hot air.

During the sampling process, 300 g of cacao beans were extracted from the central portion of the cacao seeds mass in each of the six compartments of the fermentation box. These beans were specifically designated for sensory analyses. Additionally, 50 g of cacao seeds were extracted for untargeted metabolomics analyses.

Quality analysis

For the descriptive sensory analysis and evaluation of chocolate flavour profiles, chocolate bars containing 67% (w/w) cacao solids were produced. The laboratory of sensory analysis at CIRAD (Centre de Coopération Internationale en Recherche Agronomique pour le Développement, Paris, France) employed a standardized protocol.¹⁸ The descriptive sensory analysis was conducted using the consensus profile method outlined by the International Organization for Standardization (ISO).⁴⁰

The blind tasting sessions involved 8 trained judges who evaluated the chocolate samples in duplicate. Two evaluation sessions took place. In the initial session, the judges agreed upon the sensory attributes to be assessed. Subsequently, each judge individually evaluated all the chocolate samples, assigning scores on a scale from 0 to 10 for the predetermined sensory attributes. Cronbach's alpha coefficient was used to assess the reproducibility of the judges' evaluations.

To confirm the assumptions of normality and homogeneity of variance, Shapiro–Wilk and Levene's tests were conducted. Because these assumptions were not met, a non-parametric test, specifically the Kruskal–Wallis test, was employed to determine significant differences among the chocolate samples from the various geographic zones. A Bonferroni correction was applied to post-hoc test pairwise comparison zones.

Untargeted metabolomic analysis: the chemical fingerprint of fermented and dried cacao beans

Extraction, instrumental analysis, and data processing procedures followed established methodologies.^{19,41} Initially, all samples were ground and defatted. An extraction solution composed of acetone water:formic acid (70:29.5:0.5) was employed, and ultrasonication was utilized during the extraction process.

For instrumental analysis, an ultra-performance liquid chromatography-electrospray ionization (UPLC-ESI)/quadrupole-time-of-flight-mass spectrometry (MS) system (Acquity; Waters Inc., Milford, MA, USA) was employed. Chromatographic separation was achieved using a UPLC CSH C18 column (1.7 µm particle size, 2.1 mm × 100 mm; Waters Inc.) coupled with a pre-column

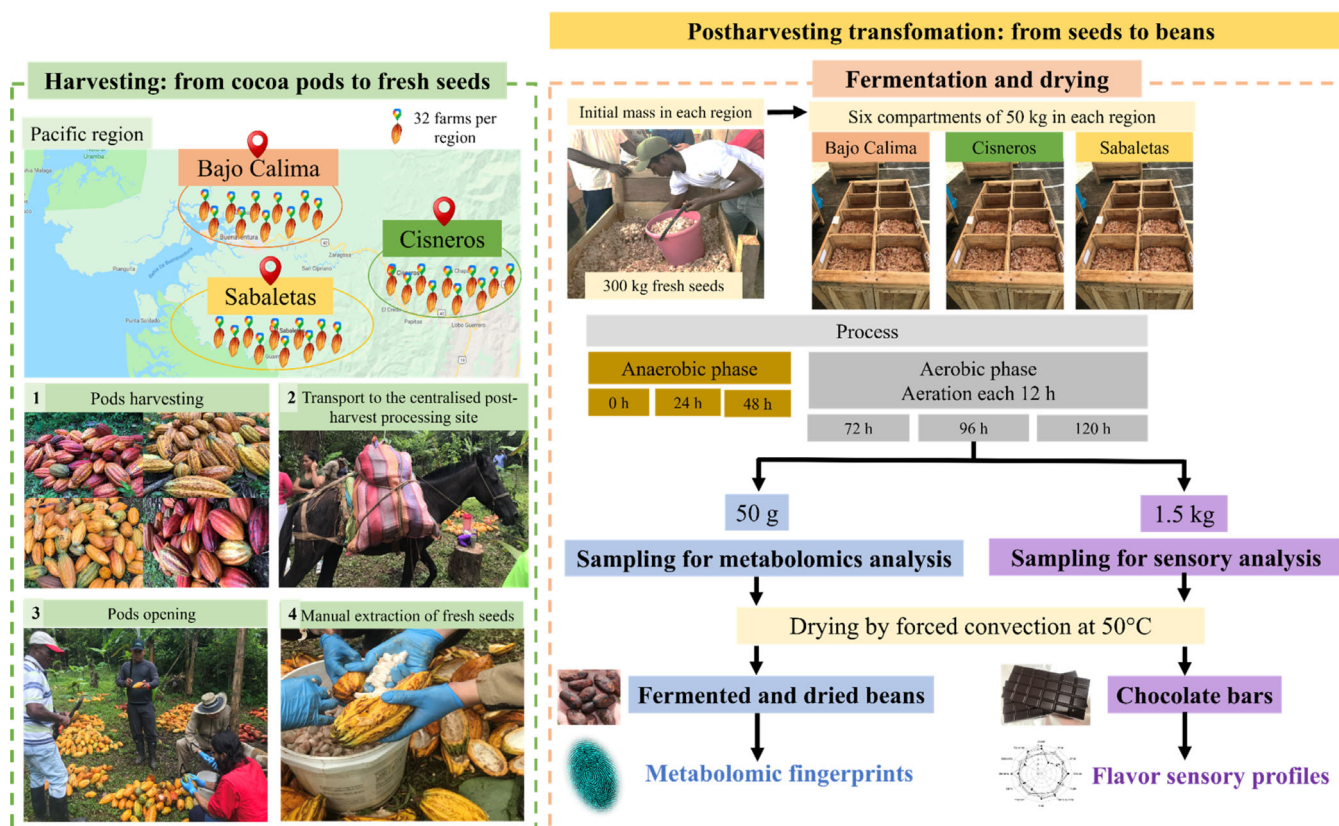


Figure 3. Scheme of the methodology developed to analyze the quality potential of the cacao cultivated in the selected farms from Sabaletas, Cisneros and Bajo Calima.

VanGuard CSH C18 (1.7 μ m particle size, 2.1 mm $\times$ 5 mm; Waters Inc.). A gradient elution of water and acetonitrile (solvent B) was utilized. ESI-MS analyses were conducted in positive ionization mode, employing a full scan range of 100 to 1500 Da in continuous mode. To ensure quality control, two pooled samples were measured after every 10 samples.

Data processing involved the use MassLynx, version 3.1 SCN 639 (Waters Inc.) and Progenesis Q1 software (Waters Inc.) to select and align chromatographic peaks. The effect of batch-related drift on the quality control baseline was corrected using Metabodrift.⁴² Data normalization was performed using the LOESS (i.e. locally reweighted scatterplot smoothing) method ($\alpha = 0.5$). Finally, the data were transformed using a dilution factor that accounts for the mass of the cacao powder and the mass of the sample upon resuspension.

To integrate the extensive metabolomic data and conduct multivariate statistical analysis, a PCA was employed to visualize the data patterns and trends. Subsequently, a PERMANOVA was conducted to identify significant differences in the metabolomes of the fermented and dried beans between the three zones. Additionally, partial least square discriminant analysis (PLS-DA) was utilized to evaluate the relationship between the metabolomes and the different zones, as well as to assess the predictive ability of the classification model.

Microorganisms (bacteria)

The bacterial cells associated with the cacao bean mass were separated from the samples using the protocol previously described.⁴³ Briefly, frozen beans (20 g) were homogenized with

0.85 g per 100 mL of NaCl in 250-mL Erlenmeyer flasks using a rotatory shaker (Thermo Scientific MaxQ 4000; Invitrogen, Burlington, ON, Canada) at 150 rpm for 30 min. The resulting fluid was decanted and filtered through sterile gauze. The free-pulp solution was then centrifuged at 3220 $\times g$ at 12°C for 20 min (Hermle, Z32HK; Labortechnik GmbH, Wehingen, Germany) and the pellet obtained was resuspended in 10 mL of sterile saline solution. DNA extraction from this solution was performed using an UltraClean Microbial DNA isolation kit (MoBio, Carlsbad, CA, USA). The concentration and purity of the extracted DNA were quantified using a NanoDrop 1000 Spectrophotometer (Thermo Fisher Scientific, DE, USA) and stored at -20°C .

The bacterial communities associated with the cacao bean mass were analyzed by sequencing the hypervariable V4 region of the 16S rRNA gene. Bacterial libraries were prepared following the protocol previously described⁴⁴ which involved a two-step PCR procedure. In the first PCR, modified primers 515F-806R,⁴⁵ with a linker region in the 5' end were used. In the second PCR, barcodes, as well as the Illumina i5 and i7 regions, were added to the amplicons and followed a specific protocol.⁴⁶ The libraries were sequenced using an Illumina MiSeq with the kit V2 (Illumina, San Diego, CA, USA) (2 $\times$ 250 bp).

Bioinformatic analysis was conducted to taxonomically identify the composition of the bacterial communities. First, the samples were demultiplexed using an in-house script. The quality of the sequences was assessed using FastQC v0.11.7 (<http://www.bioinformatics.babraham.ac.uk/projects/fastqc>). Adapters and linker sequences were removed from the partial sequences of the 16S rRNA gene using Cutadapt, version 1.12.⁴⁷ Low-quality

sequences, with an average Phred score < 20 and a read length < 200 nucleotides, were identified and removed from the datasets using Trimmomatic, version 0.38.⁴⁸ The Qiime 2 pipeline v_2018 (<https://forum.qiime2.org/t/qiime-2-2018-8-release-is-now-live/5860>) was employed for the analysis of the 16S rRNA gene amplicon libraries. The Dada2 (<https://qiime2.org>) algorithm was used to denoise the sequences and generate amplicon sequence variants (ASVs). The pair-end sequences were inputted into Dada2 with a truncation length of 200 bp. ASVs that were present in fewer than two samples or had less than four sequences in all the samples in the feature table were excluded from further analyses. The taxonomic assignment of the ASVs was performed using the Silva Database, version 132.⁴⁹

RESULTS AND DISCUSSION

Regionalization of GIs

The GI for the Buenaventura study region was identified as part of the cacao southwest region, specifically the 4C sub-region and central Calima province number 43 in the department of Valle del Cauca.¹¹ The climate and soil temperature conditions in this province are characteristic of the lowland rainforests in the Pacific region. However, the GI for zones 2 and 3 differs from that observed for zone 1 as a result of their lower elevation. Soil temperature seasonality is lower in zones 2 and 3 compared to zone 1.

The local level geospatial PCA analysis demonstrated that each sampling zone possesses specific climate characteristics (Fig. 4). Consequently, the study zones have distinct climate GIs, as indicated in Table 1, which are further supported by the national level regionalization. Solar radiation and daily temperature range emerge as the primary driving variables in zone 1, suggesting fewer extreme conditions compared to the other zones. Conversely, zone 2 is strongly influenced by high levels of rainfall and humidity. Temperature exhibits the most significant effect on zone 3.

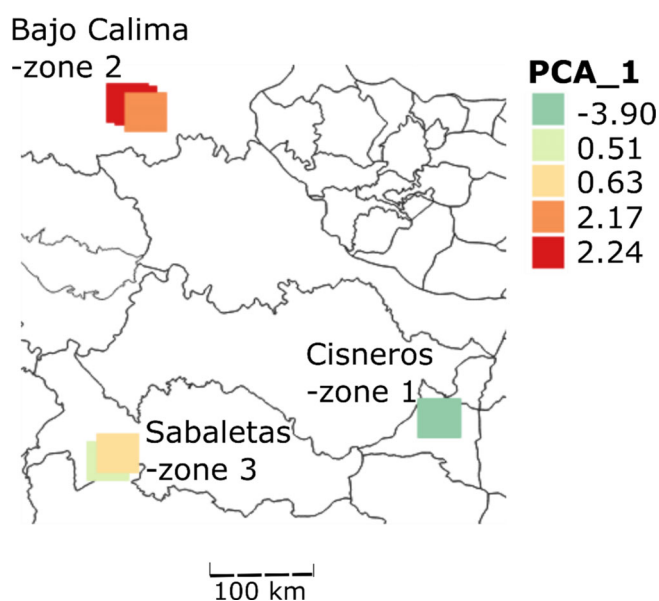


Figure 4. Local level climatic zones determined by a geospatial analysis of climate mapped as a principal component analysis of axis 1 (% explained variance by the principal component).

Agronomic baseline

The types of cacao found in the whole study region were FQ-1, FEAR5 (Trinitario), F61, SUI-83, SUI-62, GS, EET-72, TSH-565 (Trinitario), ICS-95, CCN-51, FCH-8, FQ-1 and FEE-2. In zones 3 and 2, the main cacao cultivars grown were ICS 95 (Trinitario x Criollo) and CCN 51, which were planted in the context of a rural development project aimed at encouraging cacao cultivation. Most farms in the region were small-scale, ranging from 1 to 4 ha in size. The average age of the cacao plantations was approximately 9 years, but most farmers did not maintain records of the exact number of cacao trees on their farms. Excessive shading was observed in certain farms in zones 2 and 3, likely as a result of the efforts to mitigate the effects of high rainfall. This shading can affect cacao production. Diseases such as witch's broom, frosty pod rot and black-pod rot also require intensive agronomic management to mitigate their impact on the plantations. The three study zones featured diversified farming systems, with a focus not only on cash crops, but also on other native crops for domestic consumption. The most common were peach palm (*Bactris gasipaes* Kunth), coconut (*Cocos nucifera* L.), cassava (*Manihot esculenta* Crantz), citrus fruits (*Citrus spp.*), borojó (*Boroja patinoi* Cuatrecasas) and Chinese potato (*Coleus parviflorus* L.). Some farmers also mentioned the presence of timber trees such as guino (*Carapa guianensis* Aubl.) and sajo (*Campnosperma panamense* Standl.).

Pests and diseases

In total, 1358 cacao pods were assessed for pest and disease damage. Regarding pests, rodent-damaged pods were only recorded in a single plantation in zone 2, with two pods (1.5% damage) so they were not statistically analyzed. In the overall study region of Buenaventura, geometrid and sucking insect damage (excluding cacao mirids) were the most commonly observed on cacao pods, accounting for $19.2 \pm 1.9\%$ and $17.9 \pm 1.9\%$ of the pods, respectively (mean $\pm$ SE). Pod borers and signs of cacao mirid feeding activity were much less frequent, present in only $3.5 \pm 0.8\%$ and $3.4 \pm 0.3\%$ of the pods, respectively.

Zone 1 had higher infestation levels of cacao mirids and pod borers compared to the other two zones (Fig. 5A,D). No pods damaged by these pests, nor the insects themselves, were observed in zone 2. However, zone 2 recorded the highest levels of fruits damaged by geometrids compared to the other zones (Fig. 5B). In these plantations, a large number of geometrid larvae were observed feeding on cacao foliage (M. Porcel, personal observation). Zone 2 also had a high number of pods with signs of sucking activity, presumably from membracids. Zone 3 showed high activity of geometrids and membracids (Fig. 5B,C), whereas a moderate number of pods damaged by cacao mirids and pod borers were observed (Fig. 5A,D).

Pest damage was similar between zones 2 and 3 but differed significantly from zone 1, which showed higher damage caused by *Monalonion* spp. and cacao pod borers, which are well-established pest species in the country.⁵⁰ These differences in pest damage could be attributed to the climatic conditions determined by altitude (Fig. 4). Previous studies on cacao pests in other regions of the country have primarily focused on inland cacao plantations that have climatic conditions more similar to zone 1. Therefore, the high levels of geometrid and sucking insect damage (mainly caused by the membracid *Horiola picta* Coquebert) could be associated with the high humidity and temperatures found in these zones. Under such conditions, there may be a higher turnover of cacao damage, particularly in the case of geometrids, which can cause severe deformations to

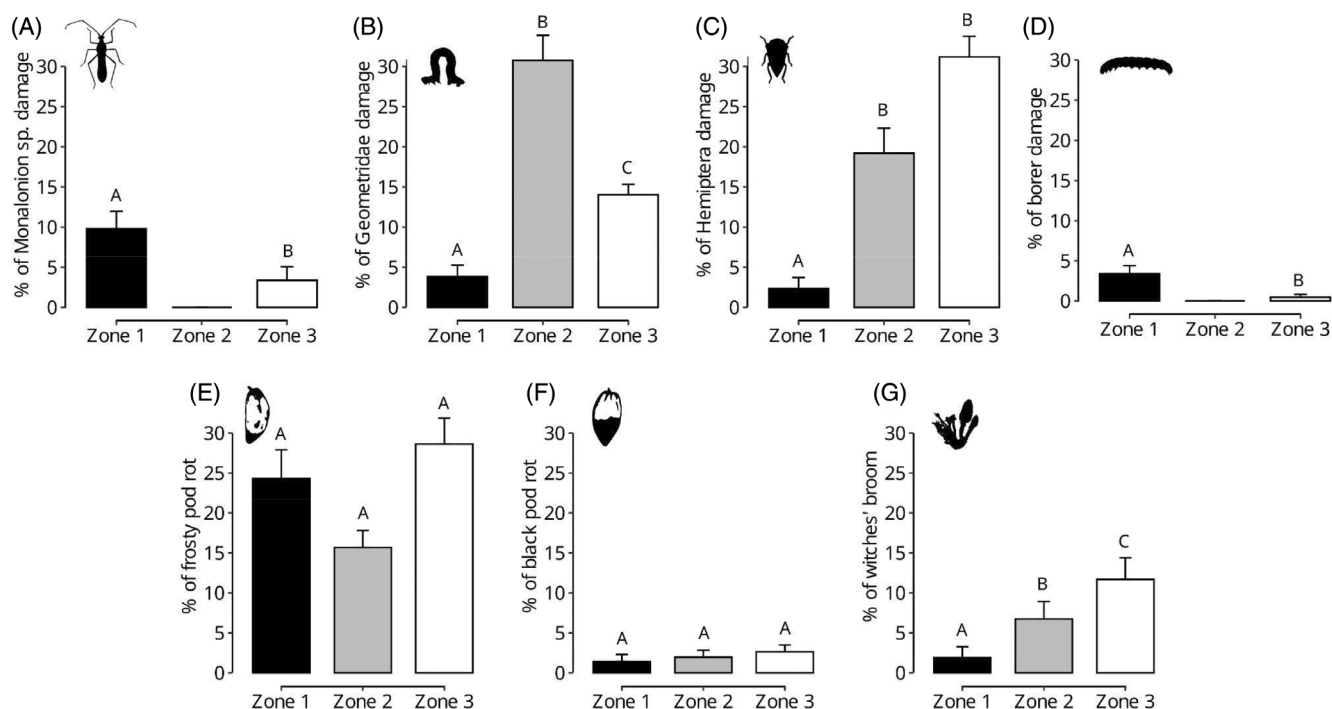


Figure 5. Proportion of cacao pods (mean $\pm$ SE) with feeding damage caused by (A) cacao mirids, (B) geometrid larvae, (C) sucking insects (other than mirids) and (D) lepidopteran pod borers and incidence per pod of (E) frosty pod rot disease, (F) black pod rot disease and (G) witches' broom disease. Different letters indicate statistically significant differences (GLMM: Tukey's test, $P < 0.100$). No letter is assigned when the percentage is zero.

pod and consume entire cherelles. The cacao mirid species identified in the study region was *Monalonion dissimulatum* Dist, which is the most common species associated with cacao in the neotropics.⁵¹ All the pod borers identified belonged to the *Ecdyolopha* genus, whereas species of the *Carmenta* genus were not found in the present study.

As part of the participatory research approach, it was observed that, although farmers were able to identify unhealthy cacao pods, they struggled to determine the specific type of pest or disease affecting the plants. There was significant variation in the way farmers managed unhealthy plants in the field. Some farmers mentioned following management methods recommended by agrochemical dealers, whereas others relied on traditional approaches for disease management such as sanitary harvesting. Additionally, important agronomic practices such as pruning and removing cacao pods affected by pathogens were not commonly practiced in the region, leading to a continuous presence of diseased pods and facilitating the spread of pathogens. Cacao growers expressed that the implementation of more rigorous agronomic practices increased production costs, thereby reducing profitability. However, a few cases of farmers with better management practices were also observed.

FPR was identified as the most significant disease affecting the cacao production system (Fig. 5E). In mature pods (2.5–3.5 months old), brown spots with irregular edges were observed. The severity of the disease varied, but many infected pods were in the sporulation phase, which is critical for the optimal spread of spores by the pathogen. This finding indicates that infected pods attached to the trees were not effectively removed. In total, 33 isolates were obtained from three study zones. Macroscopic characteristics confirmed that the isolates collected corresponded to *Moniliophthora* spp. for FPR and WBD. However, it was not possible to isolate *Phytophthora* spp. from cacao pods affected by BPR.

Previous reports have already highlighted that FPR disease is one of the main factors limiting yields in tropical America.^{52–53} It is twice as destructive as BPR and more challenging and dangerous to control compared to WBD. This poses a serious challenge for cacao growers in the region, emphasizing the urgency to develop new cacao materials that are resistant to FPR.

Genetics

Phylogenetic analysis

The maximum likelihood phylogenetic analysis resulted in a tree with low bootstrap values, indicating insufficient support for the tree nodes. The samples from Buenaventura were distributed across different branches, suggesting that they likely have diverse genetic backgrounds and belong to distinct cacao genetic groups (Fig. 6). Farms in zone 1 exhibited materials related to those from the Colombian Cacao Federation (Fedecacao), including FQ-1 and FEAR5, as well as Colombian hybrid commercial materials such as F61, SUI-83, SUI-62 and GS. One tree in zone 1 was potentially associated with Ecuadorian cacao EET-72. Materials from zone 2 showed close relationships with Ecuadorian materials (EET), the Trinidad commercial material TSH-565 and materials from Fedecacao. In zone 3, the materials were related to ICS-95 and CCN-51. Other cultivated materials demonstrated stronger connections with Fedecacao, such as FCH-8, FQ-1, and FEE-2, whereas one material was linked to cacaos from Tumaco (Colombian Pacific coast region). The PCA results (Fig. 7) indicated that the samples did not form distinct clusters based on their zones, and the PERMANOVA test confirmed no significant genetic differentiation between the zones ($P = 0.19$).

In the supervised mode of population structure analysis, we were able to determine the ancestry of the samples from Buenaventura (Fig. 7). Our findings suggest that the materials found in this region are likely the product of admixture. All the genotypes

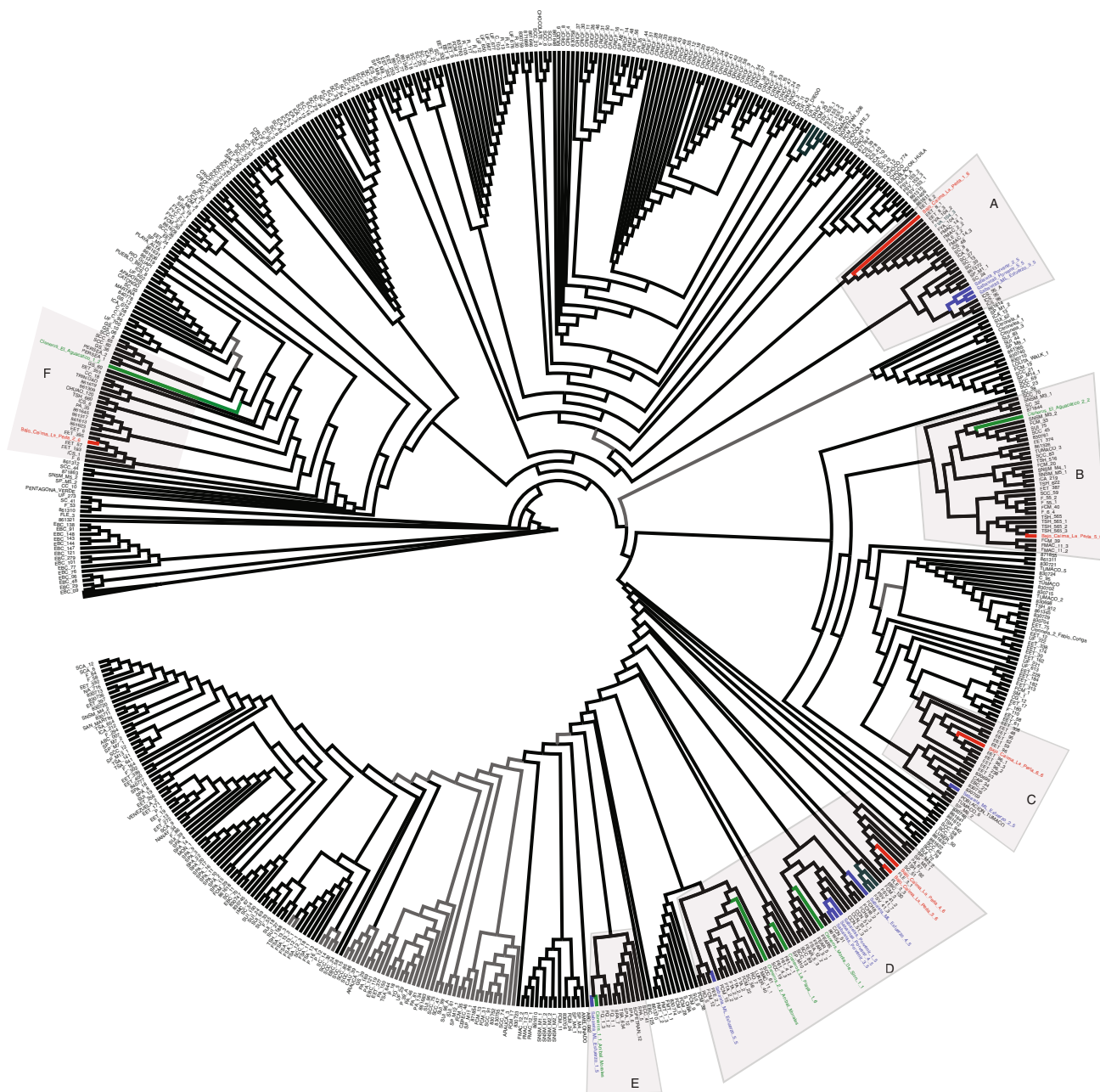


Figure 6. Maximum likelihood phylogenetic tree showing the relationships of the samples from Buenaventura (green, red and blue labels) with AGROSAVIA cacao germplasm.

exhibit mixed ancestry from the Amelonado and Criollo genetic groups, with the majority also displaying ancestry from the Iquitos and Contamana genetic groups. Only two materials, one from zone 2 and one from zone 3, showed apparent ancestry from the Nacional genetic group. These results indicate that the cultivated genotypes in these farms have a hybrid origin, combining genetic traits from different cacao genetic groups.

Quality analysis

Untargeted metabolomic analysis to define a chemical fingerprint of cacao

Metabolomics serves as a valuable tool for evaluating the impact of various factors, including geographical origin and postharvest processing stages, on the biochemistry of cacao beans. In the present

study, an integrated approach combining untargeted metabolomics and multivariate statistics was employed to analyze the chemical profiles that define the unique characteristics of cacao products from different agroecological regions in Buenaventura.

After processing the raw files, approximately 4400 features were detected within the mass range of 100–1500 Da in positive mode. The unsupervised PCA revealed distinct clustering of metabolomic fingerprints among the regions (Fig. 8A). The majority of the variance was explained by the first two components (Fig. 8B), with notable differentiation observed between zone 1 and zones 2 and 3, as confirmed by the significant differences identified through the PERMANOVA analysis ($P < 0.001$). Each metabolome represents a unique chemical fingerprint of cacao from this specific area of Colombia, as validated by the adjusted PLS-DA model.

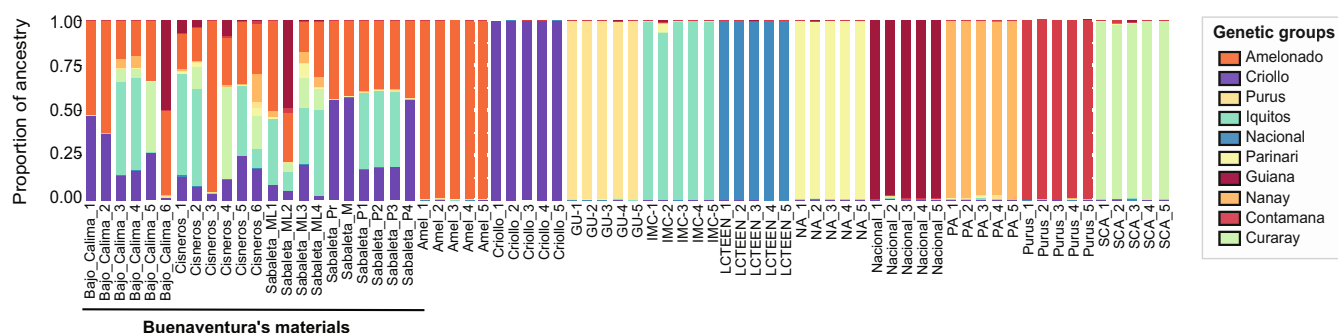


Figure 7. PCA of the population structure of the *Theobroma cacao* materials using single nucleotide polymorphisms (SNPs). Colors in each bar correspond to the probability of belonging to a reference genetic group. The pure clusters on the right side correspond to the reference samples.

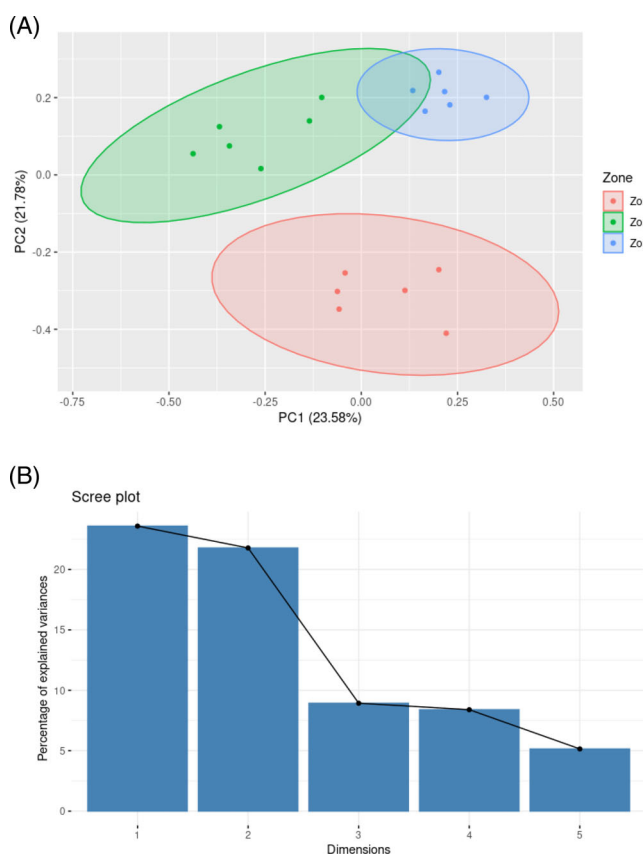


Figure 8. (A) PCA score plot of the metabolomic fingerprints of the fermented and dried cacao samples from each zone; (B) Scree plot of the variance percentage explained by each component. The first two PCs explained 45.4% of the total variance.

The model demonstrated high predictive power, with a cross-validation using the leave one out algorithm (LOOCV) yielding a predicted residual sum of squares (Q₂) of 0.884. Moreover, the individual component R² indicated robust model fitting, explaining 0.962 of the variance.

These chemical fingerprints consist of low molecular weight metabolites known as chemical flavor precursors, which play a crucial role in the non-enzymatic browning reactions during subsequent stages, such as roasting (see Supporting information, Fig. S2). Optimal conditions of temperature, time and water activity facilitate the interaction of these metabolites in Maillard reactions, leading to the generation of volatile aroma compounds

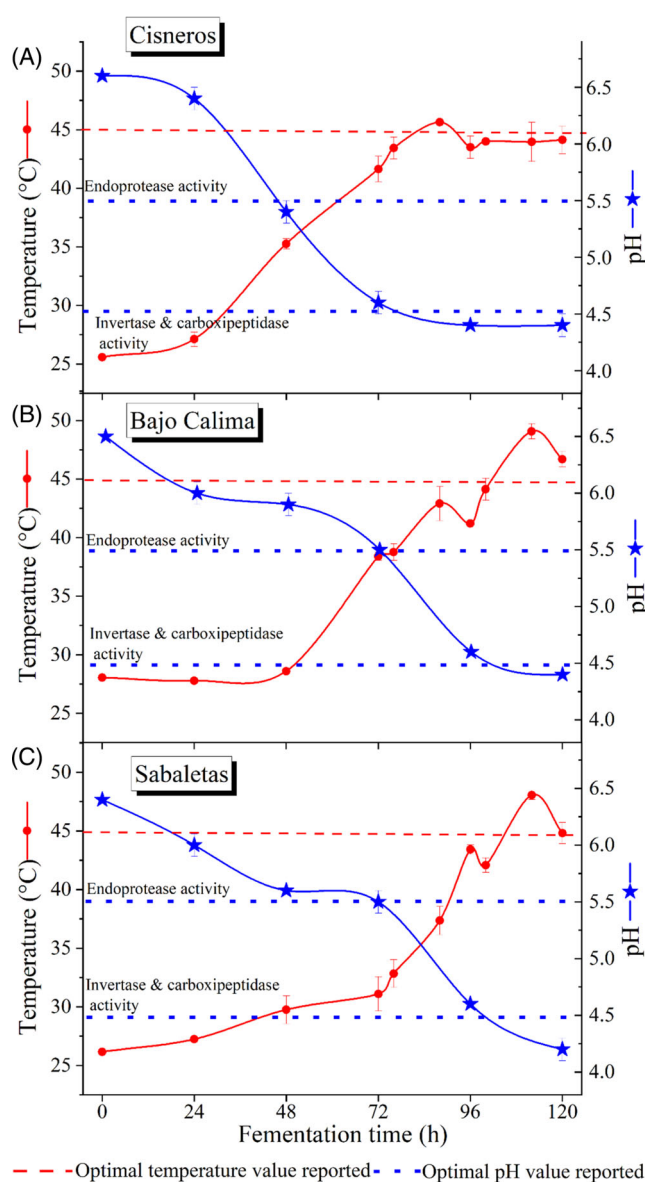


Figure 9. Dynamics of temperature (red line) and pH (blue line) during the cacao fermentation process in the three zones: (A) zone 1, (B) zone 2 and (C) zone 3.

associated with distinct sensory descriptors of cacao, including fine, bulk or undesirable flavors. Additionally, non-volatile

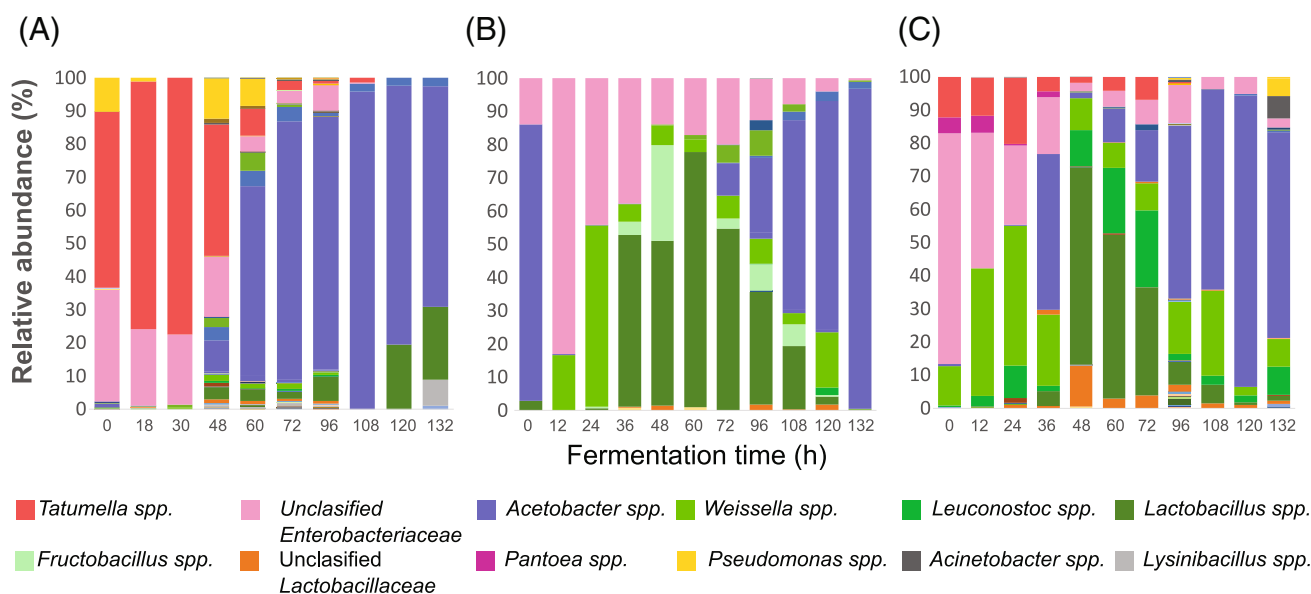
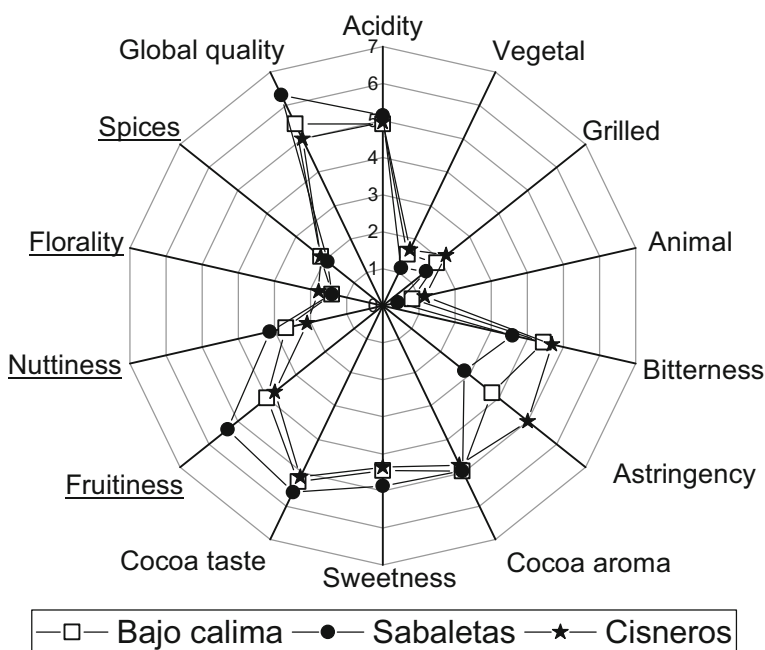


Figure 10. Bacterial succession during cacao bean fermentation. The relative composition of the bacterial communities throughout the cacao fermentation process is shown for (A) zone 1, (B) zone 2 and (C) zone.



Sensory attribute	Cisneros	Bajo calima	Sabaletas
Fine flavor			
Fresh Fruits	ab	ab	c
Nuts	a	a	a
Floral	a	a	a
Spices	a	a	a
Global quality	ab	ab	c
Basic flavor			
Sweet	a	a	a
Sour	a	a	a
Bitter	a	a	a
Astringent	a	b	c
Cocoa taste	a	a	a

Basic flavors atributes Fine flavors atributes

Figure 11. Sensory attributes of chocolates produced with cacao beans from zone 1 (Cisneros), zone 2 (Bajo Calima) and zone 3 (Sabaletas). Different letters indicate significant differences (ANOVA, Tukey's test, $P < 0.050$).

compounds such as phenolic compounds and methylxanthines (theobromine, caffeine and theophylline) contribute to defining the flavor profile and overall quality of cacao.

Fermentation variables play a crucial role in the cacao seed's mass and heat transfer processes, which are closely tied to the biochemical changes occurring within the seed. These changes involve a combination of enzymatic and non-enzymatic reactions

that contribute to the formation of flavor precursor metabolites. To monitor these processes, important physicochemical variables such as internal cacao mass temperature and internal pH of the cacao beans were measured throughout the fermentation process (Fig. 9).

Overall, an increase in temperature was observed in all regions, albeit with slight variations in the trend. Zone 1 exhibited a

significant upward trend starting at 24 h, whereas, in zones 2 and 3, the temperature increase began after 48 and 72 h, respectively. Beyond the 72-h mark, a consistent progressive temperature rise was observed in all regions, eventually reaching approximately 45°C. This temperature increase is attributed to the exothermic reactions caused by acetic acid bacteria thriving in the aerobic conditions generated by the aeration of the fermentation mass.

A concurrent sharp decrease in pH was observed throughout the fermentation process across all zones. The pH levels ranged from 6.5 to 4.5. These pH conditions in all regions facilitate the optimal functioning of enzymatic processes and biochemical changes within the cacao seeds. Enzymes such as proteases, invertases and polyphenol oxidases effectively act on proteins, carbohydrates and phenolic compounds at the ideal temperature and pH, leading to the generation of lower molecular weight compounds known as flavor precursor metabolites in cacao. These metabolites include free amino acids, peptides and reducing sugars, all of which contribute to the development of unique flavors in cacao.

Microorganisms

Bacterial diversity

Cacao bean fermentation is a vital process that plays a key role in the development of chocolate (or cocoa) flavor by generating essential chemical precursors. During fermentation, both bacteria and fungi undergo physical and biochemical transformations of the bean structures. This transformation follows a predictable microbial succession, involving yeast and various functional groups of bacteria. Specifically, the bacterial succession starts with organisms related to the Enterobacteriaceae (ENT group), which then transition to lactic acid bacteria (LAB group) and finally to acetic acid bacteria (AAB group).⁵⁴ The dynamics of microbial communities involved in cacao bean fermentation can now be assessed using high-throughput sequencing of phylogenetic markers, a method known as metataxonomy. This approach overcomes the limitations of traditional cultivation-based methods and provides a more detailed understanding of the microbial composition involved in the fermentation process.⁵⁵

The fermentation processes in the three study zones were examined using 16S rRNA metataxonomy (Fig. 10). Although all three microbial groups were observed in all fermentations, there were differences in the fine-level composition, such as genus and species, as well as relative abundance, between the fermentations. For example, *Tatumella* was present during the initial fermentation steps in zones 1 and 3 but not in zone 2, whereas *Pantoea* was only found in zone 3. Another unexpected difference was the presence of *Pseudomonas* in zone 1, a bacterial taxonomic group that is not typically found in fermentations. Perhaps the most notable difference between the fermentations was the atypical microbial succession observed in the fermentation from zone 1, where it transitioned directly from ENT to AAB, bypassing the LAB group. This faster transition to AAB might be responsible for the rapid increase in fermentation temperature and drop in pH observed at 48 h (Fig. 10) because the production of acetic acid by AAB is an exergonic reaction. This accelerated transition may impact the transformation of the pulp and beans during fermentation and could explain the differences in the metabolomics profile observed between zone 1 and the other two locations.

Although recent studies have not found a clear relationship between the absence of lactic acid bacteria (LAB) in cacao fermentation and distinct characteristics in the organoleptic profile of chocolate, it is still possible that LAB play a role in the fine transformation of pulp and contribute to the development of sweet and

distinct cacao notes. LAB has been shown to transform citric acid during the initial stages of fermentation, preventing its transfer into the beans, and also reduces fructose to mannitol, which may diffuse into the beans.^{56–57} Recent evidence has demonstrated the accumulation of mannitol during the fermentation process. These transformations can lead to additional processes that contribute to the cacao flavor, beyond those caused by acetic acid alone.⁵⁸

Sensory profile of chocolate and special quality

Chocolates made from the cacao beans of the three sampling zones exhibited distinct sensory attributes, including fresh fruits, floral notes, spices and nuts (Fig. 11). These unique attributes classify the cacao from this region as a fine flavor product, according to the International Cocoa Organization.⁵⁹ Producers who are able to obtain higher quality cacao beans have greater opportunities to access different market segments and achieve better profitability in their sales.^{39,54}

Although there were differences in climate and genetic composition of the plantations, as well as variations within each fermentation batch per zone, we did not find significant differences in cacao bean quality among the zones. Some minor variations in the intensity of certain attributes, such as fresh fruits, overall quality and astringency, were observed (Fig. 11); however, there were no distinct patterns in the types of attributes that formed the sensory profiles of each region.

Thus, there is no direct correlation between climate and the cacao varieties cultivated in these regions that determine the production of fine flavor cacao beans. Buenaventura demonstrates a combination of natural characteristics that contribute to the production of high-quality cacao products, with a unique and specialty quality associated with fruity-aroma chocolates.

CONCLUSIONS

Colombia boasts a rich diversity of cultivated cacao throughout the country, which is preserved in *ex situ* collections such as the national germplasm collection conserved at AGROSAVIA research stations.³¹ This genetic reservoir plays a crucial role in the development of high-quality cacao in the country but, to enhance this diversity, methodologies for DOs need to be implemented.

In the present study, we propose an evidence-based framework for supporting establishing DOs of cacao. Our analysis indicates specific GIs for each of the three sampling zones, with Buenaventura identified as a region with natural characteristics to produce fine flavour cocoa products. Each zone exhibited a high differentiation and diversity of cacao cultivars. Buenaventura has the potential to be designated as a future DO for cacao from the Pacific region of Colombia, characterized by its unique fruity-aroma chocolates. However, each study zone exhibited specific metabolomic characteristics influenced by the unique climatic conditions of each zone. Additionally, distinct pest species and genetic footprints were identified across the zones. Overall, this proposed framework highlights the potential for supporting GIs and DOs in cacao. The integration of scientific expertise and community collaboration was vital in achieving a comprehensive understanding of the cacao quality and characteristics specific to Buenaventura.

AUTHOR CONTRIBUTIONS

CEG-O and SE conceived the idea. CEG-O, SE, MP, RY, CR, MS, MZ, AC-Q, MS and JR collected, cleaned and analyzed the data. CEG-O

and SE led the writing and revision of the manuscript. All authors contributed critically to the drafts and approved the final version of the manuscript submitted for publication.

ACKNOWLEDGMENTS

We thank the Corporación Colombiana de Investigación Agropecuaria (AGROSAVIA) for providing funding. We gratefully acknowledge the generous assistance provided by cacao farmers associations of Buenaventura: Asociación de Productores Campesinos de Cisneros, Asociación de Agricultores y Cacaoteros de Bajo Calima, and Asociación de Productores y Transformadores de Zacarías-Sabaletas. We also thank the University of the Andes, especially Professor Bart Van Hoof, Maite Rosales, Juanita Duque and Jesús Riascos, as well as the researchers from AGROSAVIA, Darwin Martínez, Eliseo Polanco and Rafael Novoa. Bancoldex and the Buenaventura Chamber of Commerce are acknowledged for contributing resources for the development of this research project.

CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

REFERENCES

- Marie-Vivien D and Casabianca F, Geographical indications: protection of a name or a logo? A risky shift, in *Actes de la conférence Perspectives mondiales sur les Indicateurs Géographiques*. FAO; CIRAD, Montpellier, pp. 31–33. Perspectives mondiales sur les indications géographiques, 2022-07-05/2022-07-08, Montpellier (France) (2022).
- The World Intellectual Property Organization (WIPO) 2022. https://www.wipo.int/geo_indications/en/. Last accessed 27_09_2022
- The World Intellectual Property Organization (WIPO) Geographical Indicators: an introduction. https://www.wipo.int/edocs/pubdocs/en/wipo_pub_952_2021.pdf. (2021). Last accessed 27_09_2022.
- Institute for the National Denomination of Origin (INAO), Guide du demandeur d'une appellation d'origine protégée (AOP) ou d'une indication géographique protégée (IGP) à l'exception des vins, boissons alcoolisées et boissons spiritueuses <https://www.inao.gouv.fr/eng/Official-signs-identifying-quality-and-origin/PDO-AOC> (2017).
- Fouet O, Looor Solorzano RG, Rhoné B et al., Collection of native L. accessions from the Ecuadorian Amazon highlights a hotspot of cocoa diversity. *Plants People Planet* **4**:618–637 (2022).
- González Jiménez E, Denominación de origen Cacao Chuao (fao.org). FAO-ILCA consultancy (2007).
- Quingaísa, E, Riveros H, Estudio de Caso: Denominación de Origen "Cacao Arriba" (fao.org). (2007).
- Quintana Lombeida MD and Aguilar Herrera JV, Denominación de origen de cacao ecuatoriano: ¿Un aporte de marketing global? *Innova* **3**:68–76 (2018).
- SEGOB, Declaración General de Protección de la denominación de origen Cacao Grijalva (2016), Mario Pomares: La denominación de Origen garantiza la calidad de la Granada Mollar de Elche y la defiende frente a terceros.
- de Melo A, de Tarso P, de Souza S, de Melo C, da Conceição S and Ribeiro A, Tomé-açu cacao: the importance of geographical indication for products commercialized in the international market. *Revista Ingi* **4**:1033–1047 (2020).
- González-Orozco CE and Pesca A, Regionalization of Cacao (*Theobroma cacao* L.) in Colombia. *Front Sustainable Food Syst* **6**:1–11 (2022).
- Pineda-Suarez P, Hacia una mejor coordinación interinstitucional para una operación eficiente y completa de las entidades delegadas de las DOP en Colombia. *Perspectivas mundiales de las indicaciones geográficas* Montpellier, Francia, 5–8 Julio (2022).
- Superintendencia de Industria y Comercio. SIC, <https://bit.ly/3BIQuZW> Colombia Superintendencia de Industria y Comercio. Denominaciones de Origen y Marcas no tradicionales. *Documento Tecnico*:1–76 <https://bit.ly/3dMW7OK>. Colombia. https://www.sic.gov.co/sites/default/files/Denominaciones_Origen_Marcas_Tradicionales%20%20.pdf
- Oberthür T, Läderach P, Posada H, Fisher MJ, Samper LF, Illera J et al., Regional relationships between inherent coffee quality and growing environment for denomination of origin labels in Nariño and Cauca, Colombia. *Food Policy* **36**:783–794 (2011).
- Decazy F, Avelino J, Guyot B, Perriot JJ, Pineda C and Cilas C, Quality of different Honduran coffees in relation to several environments. *J Food Sci* **68**:2356–2361 (2003).
- Unidad de Planificación Rural Agropecuaria (UPRA) <https://www.upra.gov.co/uso-y-adequacion-de-tierras/evaluacion-de-tierras/zonificacion> Colombia, 2021. www.dof.gob.mx/nota_detalle.php?codigo=5449991&fecha=29/08/2016#gsc.tab=0. Mexico
- Hernandez CE and Granados L, Quality differentiation of cacao beans: implications for geographical indications. *J Sci Food Agric* **101**:3993–4002 (2021).
- Escobar S, Santander M, Vaillant F, Zuluaga M and Rodríguez J, Fine cacao beans production: tracking aroma precursors through a comprehensive analysis of flavour attributes formation. *Food Chem* **365**: 1–11 (2021).
- Santander M, Vaillant F, Sinuco D, Rodríguez J and Escobar S, Enhancement of fine flavour cacao attributes under a controlled postharvest process. *Food Res Int* **143**:110236 (2021).
- González-Orozco CE, Osorio-Guarín JA and Yockteng R, Phylogenetic diversity of cacao (*Theobroma cacao* L.) genotypes in Colombia. *Plant Genet Resour: Characterization Utiliz* **20**:203–214 (2023).
- Pérez-Escobar OA, Lucas E, Jaramillo C, Monro A, Morris SK, Bogarín D et al., The origin and diversification of the hyperdiverse flora in the Chocó biogeographic region. *Front Plant Sci* **10**:1328 (2019).
- Mesa-Sánchez ÓJ and Rojo-Hernández JD, On the general circulation of the atmosphere around Colombia. *Rev Acad Colomb Cienc Ex Fis Nat* **44**:857–875 (2020).
- Fick SE and Hijmans RJ, Worldclim 2: new 1-km spatial resolution climate surfaces for global land areas. *Int J Climatol* **37**:4302–4315 (2017).
- Lembrechts J, Van den Hoogen J, Aalto J, Ashcroft M, De Frenne P, Kemppinen J et al., Global maps of soil temperature. *Global Chang Biol* **28**:3110–3144 (2021).
- González-Orozco CE, Thornhill AH, Knerr NJ, Laffan SW and Miller JT, Biogeographical regions and phytogeography of the eucalypts. *Diversity Distrib* **20**:46–58 (2014).
- Mahob RJ, Feudjio Thiomela R, Dibog L, Babin R, Fotso Toguem YG, Mahot Mahot H et al., Field evaluation of the impact of Sahlbergella singularis Haglund infestations on the productivity of different *Theobroma cacao* L. genotypes in the southern Cameroon. *J Plant Dis Prot* **126**:203–210 (2019).
- Reitan T, Nielsen, a, do not divide count data with count data; a story from pollination ecology with implications beyond. *PLoS One* **11**: e0149129 (2016).
- Cosme S, Cuevas HE, Zhang D, Oleksyk TK and Irish BM, Genetic diversity of naturalized cacao (*Theobroma cacao* L.) in Puerto Rico. *Tree Genet Genomes* **12**:88 (2016).
- Fang W, Meinhardt LW, Mischke S, Bellato CM, Motilal L and Zhang D, Accurate determination of genetic identity for a single cacao bean, using molecular markers with a nanofluidic system, ensures cacao authentication. *J Agric Food Chem* **62**:481–487 (2014).
- Ji K, Zhang D, Motilal LA, Boccara M, Lachenaud P and Meinhardt LW, Genetic diversity and parentage in farmer varieties of cacao (*Theobroma cacao* L.) from Honduras and Nicaragua as revealed by single nucleotide polymorphism (SNP) markers. *Genet Resour Crop Evol* **60**: 441–453 (2013).
- Osorio-Guarín JA, Berdugo-Cely J, Coronado RA, Zapata YP, Quintero C, Gallego-Sánchez G et al., Colombia a source of

- cacao genetic diversity as revealed by the population structure analysis of germplasm bank of *Theobroma cacao* L. *Front Plant Sci* **8**:1994 (2017).
- 32 Wang B, Motilal LA, Meinhardt LW, Yin J and Zhang D, Molecular characterization of a cacao germplasm collection maintained in Yunnan, China using single nucleotide polymorphism (SNP) markers. *Trop Plant Biol* **13**:359–370 (2020).
 - 33 Larkin MA, Blackshields G, Brown NP, Chenna R, McGettigan PA, McWilliam H *et al.*, Clustal W and Clustal X version 2.0. *Bioinformatics* **23**:2947–2948 (2007).
 - 34 Kearse M, Moir R, Wilson A, Stones-Havas S, Cheung M, Sturrock S *et al.*, Geneious basic: an integrated and extendable desktop software platform for the organization and analysis of sequence data. *Bioinformatics* **28**:1647–1649 (2012).
 - 35 Guindon S, Dufayard JF, Lefort V, Anisimova M, Hordijk W and Gascuel O, New algorithms and methods to estimate maximum-likelihood phylogenies: assessing the performance of PhyML 3.0. *Syst Biol* **59**:307–321 (2010).
 - 36 Hammer Ø, Harper DAT and Ryan PD, PAST: paleontological statistics software package for education and data analysis. *Palaeontol Electron* **4**:1–9 (2001).
 - 37 Motamayor JC and Lachenaud P, da Silva E Mota JW, Loo R, Kuhn DN, Brown JS, Schnell RJ, geographic and genetic population differentiation of the Amazonian chocolate tree (*Theobroma cacao* L.). *PLoS One* **3**:e3311 (2008).
 - 38 Alexander DH, Novembre J and Lange K, Fast model-based estimation of ancestry in unrelated individuals. *Genome Res* **19**:1655–1664 (2009).
 - 39 Escobar S, Santander M, Useche P, Contreras C and Rodríguez J, Aligning strategic objectives with research and development activities in a soft commodity sector: a technological plan for colombian cacao producers. *Agri* **10**:2–32 (2020).
 - 40 International Organization for Standardization, ISO, 13299:2016. General guidance for establishing a sensory profile. Geneva, Switzerland, Sensory analysis Methodology (2016).
 - 41 Mayorga-Gross AL, Quirós-Guerrero LM, Fourny G and Vaillant F, An untargeted metabolomic assessment of cacao beans during fermentation. *Food Res Int* **89**:901–909 (2016).
 - 42 Thonusin C, IglayReger HB, Rothberg AE, Burant CF, Charles R and Evans J, MetaboDrift, a drift-correction and normalization tool for multi-batch LC-MS metabolomics data: application to study of mixed-meal tolerance in obese and non-obese subjects. *Chromatogr A* **1523**:265–274 (2017).
 - 43 Camu N, De Winter T, Verbrugghe K, Cleenwerck I, Vandamme P, Takrama JS *et al.*, Dynamics and biodiversity of populations of lactic acid bacteria and acetic acid bacteria involved in spontaneous heap fermentation of cacao beans in Ghana. *Appl Environ Microbiol* **73**:1809–1824 (2007).
 - 44 Faith JJ, Guruge JL, Charbonneau M, Subramanian S, Seedorf H, Goodman AL *et al.*, The long-term stability of the human gut microbiota. *Science* **341**:1237439 (2013).
 - 45 Caporaso JG, Lauber CL, Walters WA, Berg-Lyons D, Lozupone CA, Turnbaugh PJ *et al.*, Global patterns of 16S rRNA diversity at a depth of millions of sequences per sample. *Proc Natl Acad Sci U S A* **108**:4516–4522 (2011).
 - 46 Pacheco-Montealegre ME, Dávila-Mora LL, Botero-Rute LM, Reyes A and Caro-Quintero A, Fine resolution analysis of microbial communities provides insights into the variability of cacao bean fermentation. *Front Microbiol* **11**:650 (2020).
 - 47 Martin M, Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnet J*, [S.I.] **17**:10–12, May 2011. ISSN 2226-6089. Available: <https://journal.embnet.org/index.php/embnetjournal/article/view/200> [11 October 2023]. <https://doi.org/10.14806/ej.17.1.200>.
 - 48 Bolger AM, Lohse M and Usadel B, Trimmomatic: A flexible trimmer for illumina sequence data. *Bioinformatics* (2014) btu170.
 - 49 Quast C, Pruesse E, Yilmaz P, Gerken J, Schweer T, Yarza P *et al.*, The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. Opens external link in new window. *Nucl Acids Res* **41**:D590–D596 (2013).
 - 50 Mújica J and Martínez D, Manejo de plagas y enfermedades que afectan al cultivo del cacao. Instituto Colombiano Agropecuario (ICA) (2015).
 - 51 Babin R, in *Pest Management in Organic Cacao: Handbook of Pest Management in Organic Farming*, ed. by Vincenzo SK. CAB International, Nairobi, Kenya pp. 502–518 (2018).
 - 52 Bailey BA, Evans HC, Phillips-Mora W, Ali SS and Meinhardt LW, Monilophthora royeri, causal agent of cacao frosty pod rot. *Mol Plant Pathol* **19**:1580–1594 (2018).
 - 53 Jaimes YY, Ribeyre F, Gonzalez C, Rojas J, Furtado EL and Cilas C, Factors affecting the dynamics of frosty pod rot in the main cocoa areas of Santander state, Colombia. *Plant Dis* **103**:1665–1673 (2019).
 - 54 Lefeber T, Janssens M, Camu N and De Vuyst L, Kinetic analysis of strains of lactic acid bacteria and acetic acid bacteria in cacao pulp simulation media toward development of a starter culture for cacao bean fermentation. *Appl Environ Microbiol* **76**:7708–7716 (2010).
 - 55 Adler P, Bolten CJ, Dohnt K, Hansen CE and Wittmann C, Core fluxome and metafluxome of lactic acid bacteria under simulated cacao pulp fermentation conditions. *Appl Environ Microbiol* **79**:5670–5681 (2013).
 - 56 De Vuyst L and Weckx S, The cacao bean fermentation process: from ecosystem analysis to starter culture development. *J Appl Microbiol* **121**:5–17 (2016).
 - 57 De Vuyst L and Leroy F, Functional role of yeasts, lactic acid bacteria and acetic acid bacteria in cacao fermentation processes. *FEMS Microbiol Rev* **44**:432–453 (2020).
 - 58 ICCO. (2017). *Fine or Flavour Cocoa*. Retrieved January 14, 2018, from <http://www.icco.org/about-cacao/fine-or-flavour-cacao.html>
 - 59 Beg MS, Ahmad S, Jan K and Bashir K, Supply chain and processing of cacao—a review. *Trends Food Sci Technol* **66**:108–116 (2017).